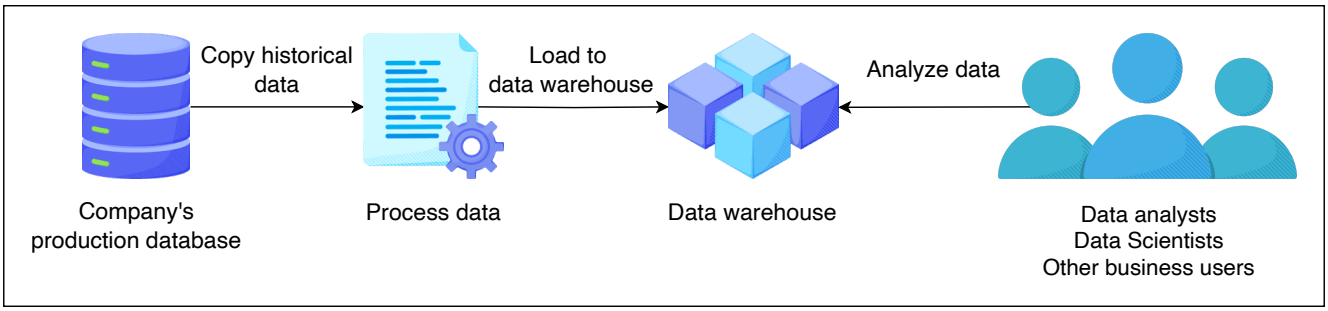
Data Warehouse

Explore the fundamental concepts of data warehouses including their purpose, architecture, and how they support ETL pipelines. Understand dimensional modeling, fact and dimension tables, and differences between data warehouses and traditional databases. Gain insights into centralized data warehouses versus data marts and see how data warehousing fits into ETL-driven business intelligence.

What is a data warehouse?

A **data warehouse** is a repository that stores historical data for analytical and BI purposes. Its main job is to copy data from the organization's internal systems, such as databases, and store it in a specific way that allows the organization to use the data to support data-driven decision-making. Data warehouses are the main repository analysts and data scientists use when pulling data for their analytics jobs.



Data warehouse copies data from the production database

Data warehouses can read and write complex queries because of a modeling approach called dimensional modeling. In **dimensional modeling**, we organize the data in the data warehouse as a series of dimensions and facts. This allows

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users to quickly and easily query the data to generate reports and perform analysis.

Star schema

A star schema is a popular design choice in dimensional modeling and data warehousing. A data repository designed as a **star schema** will have a large de-normalized table called the fact table in the middle and will be surrounded by several smaller normalized tables called dimension tables.

Fact and dimension tables

The **fact table** will store the actual historical events/data, and the **dimension tables** will store information about the columns of the fact table.

The fact table is denormalized to reduce the need for complex joins between many tables, which causes heavy load and unnecessary overhead. On the other hand, the dimension tables are normalized.

Normalizing tables means dividing them into smaller tables that store more specific data. This significantly increases consistency in the database and reduces data redundancy. However, it also reduces readability and forces the user to use join queries, which can become quite complex and cause some overhead to the system.

Example

Each row in the fact table of a bicycle shop will contain information about a specific transaction, such as the number of bikes sold, credit card information, final transaction price after discount, etc.

The dimension tables of the bicycle shop will contain information about various dimensions/columns in the fact table, such as information about the shop' products, the employees, customers, etc.

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Example of a Star Schema design of a bicycle shop in the middle earth

Data warehouses combine the advantages of relational database technology and specialized data warehousing tools and technologies to create a fast data repository suitable for handling large and complex analytical queries. One of the most common destinations for ETL pipelines is a data warehouse.

Data warehouse vs. database

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Data warehouses look like regular row-based relational databases, and sometimes it takes effort to differentiate between them. A data warehouse and a database can sometimes even use the same tool or a database vendor.

The difference between them starts with the functionality of a data warehouse compared to a traditional relational database. Usually, relational databases are row-based and store real-time data about particular parts of the business in a consistent and normalized fashion. Concretely, a normalized relational database will, at the very least, meet the requirements of 3NF (Third Normal Form).

Databases can handle massive amounts of small write queries constantly while keeping the integrity of the data and minimizing data redundancy.

Database characteristics

Comparatively, data warehouses are column-based and denormalized, optimizing them to handle large aggregation queries often used by analysts and the BI department.

Column-based databases (such as data warehouses) can efficiently retrieve aggregation queries from the disk because they physically store values from the same column tightly grouped together. Using analytical functions like sum, max, min, and count on a few columns will be much faster compared to the same scenario in a regular row-based database.

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Data warehouse characteristics

This comparison might remind us of the comparison between OLAP and OLTP systems. That's because data warehouses are an OLAP system, and transactional/production databases are an OLTP system.

Data warehouse vs. production database

Characteristics of a data warehouse

- A data warehouse is usually a column-based relational database.
- It contains copies of the original organization's data.
- It can extract data from multiple sources (both external and internal).
- Data warehouses usually store historical data for analytical purposes.
- The architecture of a data warehouse is designed for high performance for read operations but with much more data redundancy.

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- The primary purpose of the data warehouse is to provide a single source of data used for analytics and BI-related decision-making.

Centralized data warehouse vs. adding data marts

A **centralized data warehouse** is the default architecture, and it's the most simple one. It stores all the data in one place (a data warehouse). This grants users in the company one point of access to all the data available.

The second architecture approach further divides the data warehouse into several data marts.

A **data mart** is a data repository that contains data specific to a particular department, process, or use case in the organization.

Think of a data mart as a smaller and more specific type of data warehouse. Similar to how a data warehouse provides a consolidated source of historical data for the organization, so does each data mart. They offer a single source of historical data designed specifically for each department.

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Data pipeline: Loading data from production database to data warehouse to several data marts

Data marts provide several benefits, including the following:

- Data marts are specialized data stores that focus on specific business needs or departments, making accessing and analyzing relevant data easier.
- Data marts contain only a subset of the data compared to the data warehouse. So, they allow for faster querying and analysis of data.
- Data marts can help to reduce the complexity of the overall data warehouse system, making it easier to manage and maintain.

There are two types of data marts:

1. Dependent
2. Independent

A **dependent data mart** depends on internal sources within the organization. Most commonly, it draws data from the data warehouse.

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An **independent data mart** pulls data from other data sources. In addition to drawing data from the data warehouse, its a more independent version and draws

data from the organization's database.

Data warehouse and ETL

A data warehouse relies heavily on ETL pipelines to provide it with data. An ETL pipeline will usually copy data from the organization's production database and feed it to the data warehouse. It can also combine data from external sources other than the company's production database, such as IoT and sensor devices, REST API, web scraping, public data sets, etc.

Data warehousing tools

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